

Identifying the Knowledge Gap: Insufficiency of Current AI-Centric Evidence for Determining the Biological Origins of Human Emotions

Input Question

Where do human emotions originate?

Domain

Biology

Validation Status

needs_data

Problem Statement

The question seeks to identify the developmental, classification-based, and physiological mechanisms underlying the origin of human emotions, specifically distinguishing between basic mammalian emotions (e.g., fear, anger) and complex social psychological emotions (e.g., shame, guilt, pride). The provided booklet excerpt suggests emotions are constructed by the brain and influenced by learning, but lacks empirical biological evidence.

Rationale

Understanding the origin of emotions requires integrating neurobiological data with psychological theory. However, the currently allowed evidence set consists exclusively of Computer Science and Robotics literature (AI needs, data privacy, process models, human-robot interaction), which does not contain direct biological or psychological evidence regarding human emotion origins. Therefore, this report frames the inquiry as a knowledge gap, highlighting the insufficiency of current evidence to establish factual mechanisms.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: [UNGROUND_SPECULATION] Human social emotions originate from learned, needs-aware computational processes analogous to those proposed in needs-aware AI frameworks, rather than from fixed biological modules. (Note: This is a knowledge_gap placeholder; no allowed evidence supports this biological claim.)
- **Mechanism**: Abstract mechanism: Social emotions are hypothesized to emerge when a system (biological or artificial) integrates basic affective signals with learned models of 'human needs' and social context. This is purely speculative and derived by metaphorical extension from EV-Q037-bf98b0c23108f75f0b03480f (Needs-aware AI), not from biological data. No anatomical substrates are specified due to lack of evidence.

- ****Falsifiable Prediction****: If future biological data were available, one would predict that disrupting 'needs-modeling' cognitive processes should selectively impair social emotions but not basic emotions. Currently untestable with allowed evidence.
- ****Required Observations****: Neurobiological data mapping 'needs-aware' processing to specific neural circuits (currently absent in allowed evidence) ; Comparative studies distinguishing learned vs. innate emotional responses in humans (currently absent) ; Validation that AI 'needs' models are homologous to human emotional origins (currently unsupported)
- ****Risk of Being Wrong****: Extremely high. The analogy between AI needs and human emotion origins is unverified. Allowed evidence (EV-Q037-bf98b0c23108f75f0b03480f) addresses AI serving human needs, not the biological origin of those needs or emotions. This hypothesis is explicitly classified as ungrounded speculation per reviewer feedback.

Hypothesis 2

- ****Hypothesis****: [KNOWLEDGE_GAP] The origin of human emotions cannot be determined from the current allowed evidence set, as it contains only CS.AI/Robotics/Privacy sources and zero neurobiological or psychological data. A valid hypothesis requires future acquisition of domain-appropriate evidence.
- ****Mechanism****: No mechanism can be proposed. The allowed evidence IDs (EV-Q037-bf98b0c23108f75f0b03480f, EV-Q037-401acc212166e098ba65558f, EV-Q037-af406f554e9953571697f8ac, EV-Q037-f23fd384deb4ce88830af0b3) pertain exclusively to artificial intelligence, data privacy, process modeling, and human-robot interaction. None address biological emotion origins, neural substrates, or psychological construction theories. This entry serves as a formal placeholder acknowledging insufficient evidence.
- ****Falsifiable Prediction****: This hypothesis is not falsifiable in its current form because it asserts the absence of evidence rather than a testable claim. It will be replaced once relevant biological evidence is acquired.
- ****Required Observations****: Acquisition of neuroscientific literature on emotion generation (e.g., fMRI, lesion studies) ; Psychological studies on social vs. basic emotion differentiation ; Biological data validating or refuting constructionist vs. modular theories of emotion
- ****Risk of Being Wrong****: N/A. This is a meta-statement about evidence insufficiency, not a scientific claim about emotion origins. Risk applies only if future evidence contradicts the assertion that current evidence is insufficient (which is factually correct per review).

Technical Details

This is a proposed experimental template contingent on future evidence acquisition. Due to the complete absence of neurobiological or psychological data in the allowed evidence set (EV-Q037-*), no specific anatomical targets (e.g., dlPFC, amygdala) or mechanistic claims can be grounded. The design serves as a placeholder structure for a mechanism_discovery study, explicitly acknowledging that all biological premises are currently unverified knowledge gaps. The technical approach outlines a generic multimodal framework (neuroimaging + computational modeling) that would be instantiated only after acquiring domain-appropriate literature and datasets. No factual claims about emotion origins are made; this plan is strictly a methodological scaffold awaiting empirical grounding.

Datasets

Source

```json

[

"No valid biological datasets identified in allowed evidence. Current allowed evidence (EV-Q037-bf98b0c23108f75f0b03480f, EV-Q037-401acc212166e098ba65558f, EV-Q037-af406f554e9953571697f8ac, EV-Q037-f23fd384deb4ce88830af0b3) pertains to AI, privacy, and robotics, not human biology."

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Target

Future acquisition of neuroscientific datasets (e.g., fMRI, EEG, lesion studies) specifically addressing emotional processing mechanisms. Until such data is available, no target dataset can be defined.

Paper Abstract

Background: Human emotions are complex phenomena involving both basic mammalian responses and highly-developed social constructs. Understanding their origins requires robust neurobiological and psychological evidence. Methods: We reviewed the provided evidence set (EV-Q037 series) to identify mechanisms underlying emotion origins. Validation Plan: The study aims to map emotional processes to neural substrates through comparative neuroscience and neuroimaging. Results: Pending. The current evidence set consists entirely of Computer Science and Robotics literature (AI needs, data privacy, process models), which lacks direct biological relevance. Consequently, no factual claims about emotion origins can be verified. This report highlights the critical need for domain-specific biological evidence to advance this inquiry.

Methods

1. ****Pre-Study Literature Synthesis****: Mandatory review of external neurobiological literature to identify candidate brain regions and theoretical frameworks.
2. ****Hypothesis Formulation****: Develop testable hypotheses based on synthesized literature, ensuring each anatomical claim is supported by cited evidence.
3. ****Experimental Design Template****: Outline a generic protocol for comparing neural correlates of different emotional states, pending identification of specific stimuli and regions of interest.
4. ****Computational Modeling Framework****: Define abstract parameters for affective processing, to be refined once biological constraints are established.

Experiments

Baselines

```json

[

"Null Model: No specific baseline can be defined without established biological mechanisms. Current AI-related evidence (EV-Q037-bf98b0c23108f75f0b03480f) cannot serve as a biological baseline."

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Metrics

```json

[

"Evidence Grounding Score: Must reach >0.0 before proceeding to empirical metrics.",

"Literature Coverage Index: Measure of how well candidate hypotheses are supported by acquired neurobiological literature.",

"Methodological Feasibility Score: Assessment of whether proposed methods align with available future datasets."

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Ablation

Not applicable. No specific model components or brain regions have been validated for inclusion.

Validation Protocol

1. Verify that all new hypotheses are supported by at least one peer-reviewed neurobiological study.
2. Ensure no anatomical claims are made without direct evidence linkage.
3. Pre-register any future empirical studies only after the knowledge gap is filled with appropriate literature.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- ****EV-Q037-bf98b0c23108f75f0b03480f**** · arxiv · arXiv:2202.04977
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/2202.04977.pdf> · doi: Not available

- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=fb996847c2cd2dc76e59740e3c05847167990c0d4365487732616494f59689e5

- **EV-Q037-401acc212166e098ba65558f** · arxiv · arXiv:2209.09724

- authors: Not available · year: Not available

- url: <https://arxiv.org/pdf/2209.09724.pdf> · doi: Not available

- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:5|section:page-5|paragraph:1;
content_sha256=f39426f5e8aa1d9df0749f431025245098cd06874e9d8746725045f9fcae3471

- **EV-Q037-af406f554e9953571697f8ac** · arxiv · arXiv:2603.12895

- authors: Not available · year: Not available

- url: <https://arxiv.org/pdf/2603.12895.pdf> · doi: Not available

- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=f963ceb9fd93cdc0159f76b94b76b65f407e93b81c92715f7722be5024237a46

- **EV-Q037-f23fd384deb4ce88830af0b3** · arxiv · arXiv:2507.11460

- authors: Not available · year: Not available

- url: <https://arxiv.org/pdf/2507.11460.pdf> · doi: Not available

- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=5e8689c76121211a9609ec0ba8fe323c3d768c3e3852495e3e66a7c1f3ce4a57

Reviewer Comments

- The revision successfully addresses all critical issues from the previous iteration by explicitly classifying the hypothesis as a 'knowledge_gap' and removing all ungrounded neuroanatomical claims.

- Evidence grounding is now correctly represented: evidence_support_score is 0.0, supporting_evidence_ids are empty, and the text clearly states that allowed EvidenceCards (AI/Robotics/Privacy) do not support biological emotion origins.

- The experiment design has been appropriately reframed as a 'proposed template contingent on future evidence acquisition' rather than an executable protocol, satisfying the requirement to avoid fabricating feasibility without evidence.

- No Results fabrication detected; status remains correctly marked as pending/unexecuted.

- Reproducibility checklist and metadata accurately reflect the current state of insufficient evidence and the need for external literature synthesis before empirical work can proceed.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Explicitly label all biological claims as 'ungrounded speculation' until supported by new evidence.
- Document the absence of relevant data in allowed_evidence_ids.
- Maintain a log of required future literature searches.
- Ensure no AI/Robotics evidence is misinterpreted as biological fact.
- Version control for the experimental template, marking it as 'pending evidence acquisition'.